MD FAIYAZ RIZWEE

Scientific Administrative Assistant | BRIC-National Institute of Plant Genome Research
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PROFESSIONAL SUMMARY

M.Sc. Bioinformatics graduate (CGPA: 9.68) with hands-on experience in NGS data analysis, genomic variant analysis, RNA-Seq, and computational biology workflows on Linux, Mac, Windows and HPC environments. Experienced in developing end-to-end bioinformatics pipelines, genomic data processing workflows. Skilled in backend, software devlopment and web application development with experience applying machine learning to biological data. Interested in computational genomics, bioinformatics software engineering, and large-scale multi-omics analysis.

EDUCATION

Jamia Millia Islamia — M.Sc. Bioinformatics *2024 – 2026*
CGPA: 9.68 | New Delhi, India

B.N. College, Patna University — B.Sc. Biotechnology *2019 – 2022*
CGPA: 7.79 | Patna, India

TECHNICAL SKILLS

NGS & Genomics: RNA-Seq Analysis, Sequence Alignment, Variant Calling (SNP/INDEL), CNV Analysis, Differential Gene Expression, Genome Data Formats (FASTQ, BAM, VCF, GFF/GTF), SRA Toolkit, FastQC, HISAT2, Bowtie2, SAMtools, GATK, CNVkit, ANNOVAR, DESeq2, edgeR, limma-voom, IGV

Linux & HPC: Linux System Administration, Bash/Shell Scripting, High-Performance Computing (HPC) Clusters, SLURM Job Scheduling & Resource Management, Software/Environment Management, Git, GitHub, Conda, Docker

Programming & Machine Learning: Python, R (working knowledge), Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, Model Training & Evaluation, Classification, Regression, Deep Learning, NLP, LLMs, RAG, LLM Fine-Tuning (LoRA/PEFT), Embeddings, Vector Search, Prompt Engineering, Biological Sequence Analysis, ML for Biological Data

Web, Software & Database Development: Django, FastAPI, streamlit, React.js, REST APIs, PostgreSQL, SQLite, SQLAlchemy, HTML5, CSS3, Bootstrap

Work EXPERIENCE

BRIC – National Institute of Plant Genome Research (NIPGR) — *New Delhi, India*

Scientific Administrative Assistant | Jan 2026 – Present

- **Therapeutic Peptides:** Working on the discovery of novel therapeutic peptides with potential activity against bacterial and fungal infections. Applying sequence-based analysis and computational prediction approaches to identify and prioritize candidate antibacterial and antifungal peptides for downstream evaluation. (Ongoing)
- **Nanobody CDR-H3 Library Design:** Analyzing large-scale single-domain antibody sequencing data to optimize CDR-H3 library design by reducing the combinatorial sequence space, with plans to develop an ML model for nanobody sequence analysis and design. (Ongoing collaboration with IIT Kanpur)
- **PlantPepDB 2.0:** Developing a comprehensive plant peptide database integrating large-scale peptide sequences, functional annotations, computational predictions, sequence analysis, and interactive search and visualization capabilities. Developed backend components for biological data integration, sequence retrieval, database management, and a scientific literature RAG system for evidence-grounded biological question answering (PI Reviewing the Work).
- **SOPHIA — Short Open Reading Frame Prediction using Holistic Integrated Analysis:** Developed a plant-focused computational framework with web and Windows applications for identifying and evaluating small open reading frames (sORFs). Integrated sequence analysis, ORF extraction and translation, feature generation, and machine-learning-based coding-potential classification into an end-to-end workflow (Manuscript Writing).
- **LmiR:** Contributed to the development of a comprehensive legume microRNA resource integrating sequence, expression, processing, and target-interaction information across 16 legume species and thousands of experimental conditions.

Implemented REST API-based data access and backend integration.

- **Plant Peptide classifier:** Developed machine-learning approaches for plant peptide family classification and explored generative language models for antimicrobial peptide (AMP) sequence generation. Applied sequence-derived features and protein language-model representations to biological sequence classification and generation tasks.

Live Projects: PlantPepDB 2.0 | SOPHIA | Plant Peptide Machine Learning

INTERNSHIPS

NextGen Helper *Remote*
Bioinformatics Trainee | 2 Months

- Developed end-to-end NGS analysis pipelines for sequence alignment, variant calling, and copy number variation (CNV) analysis.
- Processed high-throughput sequencing datasets using HISAT2, Bowtie2, SAMtools, GATK, CNVkit, and SRA Toolkit on Linux-based systems.
- Performed SNP/INDEL identification and downstream variant interpretation across genomic datasets.

Jawaharlal Nehru University (JNU) - *New Delhi, India*
Prof. R. K. Brojen Singh *Summer Research Intern | 2 Months*

- Automated collection of epilepsy-associated research papers and extracted gene–disease relationships via Python-based web scraping.
- Curated and validated biological datasets to identify the top 300 candidate genes for downstream analysis.
- Constructed biological interaction networks in Cytoscape and performed network topology analysis to identify hub biomarkers.

PROJECTS

RNA-Seq Analysis Pipeline - *SRA Toolkit • FastQC • HISAT2 • SAMtools • DESeq2*

- Downloaded raw sequencing reads from NCBI SRA using SRA Toolkit and assessed read quality with FastQC.
- Aligned reads to the reference genome using HISAT2, chosen for its splice-aware alignment, correctly mapping reads across exon–exon junctions in transcript data.
- Converted SAM to BAM, then sorted and indexed alignments using SAMtools for downstream processing.
- Quantified transcript abundance and performed differential gene expression analysis using DESeq2, followed by visualization and biological interpretation of DEGs.

Variant Calling Pipeline - *SRA Toolkit • FastQC • Bowtie2 • GATK • CNVkit • ANNOVAR*

- Downloaded raw sequencing data from SRA and assessed read quality using FastQC.
- Aligned reads to the reference genome using Bowtie2, chosen over a splice-aware aligner since variant calling works on genomic DNA with no splice junctions, giving cleaner, more consistent mapping.
- Converted SAM to BAM, sorted and indexed with SAMtools, then applied GATK (Base Quality Score Recalibration and filtering) to call high-confidence SNPs and Indels.
- Integrated CNVkit for copy number/structural variant profiling and ANNOVAR for functional annotation of clinically relevant mutations.

Gene2Therapy - *React.js • FastAPI • pyDESeq2*

- Developed a full-stack platform for differential gene expression (DEG) analysis using React.js, FastAPI, and pyDESeq2.
- Automated downstream analyses: volcano plots, KEGG pathway enrichment, disease association (OpenTargets), and interactive network visualization.
- Integrated NCBI, KEGG, and OpenTargets into a unified analytical workflow for transcriptomic data.

Live Demo: gene2therapy

PUBLICATION

Rizwee MF, Ali AM, Shah SN, Shamim T, Khan A, et al. “Patient-centric care and digital health tools.” Advances in Protein Chemistry and Structural Biology, Elsevier; 2025. DOI: 10.1016/bs.pbr.2025.08.003. (Co-author).